

CONTACT INFORMATION	National Institute of Informatics 2-1-2 Hitotsubashi Chiyoda-ku Tokyo-to 101-8430 Japan	E-mail: kphalakarn@nii.ac.jp kphalakarn@gmail.com Website: https://kphalakarn.github.io/
RESEARCH INTERESTS	• <i>Probabilistic model checking</i>: quantitative properties of Markov models and games • <i>Lattice theory and fixed point computation</i>: value iteration, fixed point uniqueness • <i>Algorithm design and analysis</i>: graph-based algorithms	
EDUCATION	University of Waterloo , Ontario, Canada Ph.D., Electrical and Computer Engineering, 2019–2023 Chulalongkorn University , Bangkok, Thailand M.Eng., Computer Engineering, 2017–2019 B.Eng., Computer Engineering (First Class Honors), 2013–2017	
HONORS AND AWARDS	Best Paper Award, ICTAC 2024 University of Waterloo Faculty of Engineering Graduate Scholarship, 2023 Ripple Graduate Fellowship, 2019–2023 Chulalongkorn University Department of Computer Engineering Graduate Fellowship, 2017–2019 Chulalongkorn University Faculty of Engineering Gold Medal of Excellence, 2017 Outstanding Academic Performance Award, Engineering Institute of Thailand, 2016 First Solution Award, ACM-ICPC World Finals 2016	
ACADEMIC EXPERIENCE	National Institute of Informatics , Tokyo, Japan <i>Researcher</i> , Research Center for Mathematical Trust in Software and Systems (09.2023–present) <i>Research Intern</i> , ERATO Metamathematics for Systems Design Project (03.2019–08.2019) RWTH Aachen University , Aachen, Germany <i>Visiting Researcher</i> , Software Modeling and Verification Group (10.2025) The University of Tokyo , Tokyo, Japan <i>Research Intern</i> , Imai Laboratory, Department of Computer Science (06.2016–07.2016)	
TEACHING EXPERIENCE	University of Waterloo , Ontario, Canada <i>Teaching Assistant</i> • ECE 606 Algorithm Design and Analysis: Fall 2020 • ECE 124 Digital Circuits and Systems: Spring 2020 Chulalongkorn University , Bangkok, Thailand <i>Teaching Assistant</i> • 2110201 Computer Engineering Mathematics (Linear Algebra): Winter 2019 • 2110202 Discrete Structures and Computability (Discrete Mathematics): Fall 2018 • 2110101 Computer Programming: Winter 2015, Fall 2016, Winter 2017, Spring 2017, Fall 2017, Winter 2018, Spring 2018	

T. Sanada, Y. Montacute, **K. Phalakarn**, and I. Hasuo. “A Coalgebraic Dijkstra Algorithm,” Proc. of the 37th International Conference on Concurrency Theory (CONCUR 2026).

K. Phalakarn, S. Pruekprasert, and I. Hasuo. “Strategy Templates for Almost-Sure and Positive Winning of Stochastic Parity Games towards Permissive and Resilient Control,” Theoretical Computer Science (TCS), vol. 1057, no. 115535, 2025, pp. 1–15.

K. Phalakarn, Y.C. Tsai, and I. Hasuo. “Widest Path Games and Maximality Inheritance in Bounded Value Iteration for Stochastic Games,” Proc. of the 23rd International Symposium on Automated Technology for Verification and Analysis (ATVA 2025), pp. 109–131.

Y.C. Tsai, **K. Phalakarn**, S. Akshay, and I. Hasuo. “Chance and Mass Interpretations of Probabilities in Markov Decision Processes,” Proc. of the 36th International Conference on Concurrency Theory (CONCUR 2025), pp. 33:1–33:19.

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K. Phalakarn, V. Suppakitpaisarn, F. Rodríguez-Henríquez, and M. A. Hasan. “Vectorized and Parallel Computation of Large Smooth-Degree Isogenies using Precedence-Constrained Scheduling,” IACR Trans. on Cryptographic Hardware and Embedded Systems (TCHES), vol. 2023, issue 3, pp. 246–269.

K. Phalakarn, V. Suppakitpaisarn, and M. A. Hasan. “Speeding-Up Parallel Computation of Large Smooth-Degree Isogeny Using Precedence-Constrained Scheduling,” Proc. of the 27th Australasian Conference on Information Security and Privacy (ACISP 2022), pp. 309–331.

K. Phalakarn, V. Suppakitpaisarn, and M. A. Hasan. “Single-round Lattice-based Multisignatures,” Proc. of the 8th International Workshop on Information and Communication Security (WICS 2021), pp. 365–371.

K. Phalakarn, T. Takisaka, T. Haas, and I. Hasuo. “Widest Paths and Global Propagation in Bounded Value Iteration for Stochastic Games,” Proc. of the 32nd International Conference on Computer Aided Verification (CAV 2020), pp. 349–371.

K. Phalakarn, K. Phalakarn, and V. Suppakitpaisarn. “Optimal Representation for Right-to-Left Parallel Scalar and Multi-Scalar Point Multiplication,” International Journal of Networking and Computing (IJNC), vol. 8, no. 2, 2018, pp. 166–185.

K. Phalakarn, and A. Surarerks. “A Matrix Decomposition Method for Odd-Type Gaussian Normal Basis Multiplication,” Proc. of the 3rd International Conference on Computer and Communication Systems (ICCCS 2018), pp. 99–103.

K. Phalakarn, K. Phalakarn, and V. Suppakitpaisarn. “Optimal Representation for Right-to-Left Parallel Scalar Point Multiplication,” Proc. of the 4th International Workshop on Information and Communication Security (WICS 2017), pp. 482–488.

K. Phalakarn, and A. Surarerks. “An Analysis of Computer Programs using λ -calculus,” Proc. of the 7th International Workshop on Computer Science and Engineering (WCSE 2017), pp. 214–218.

K. Phalakarn, **K. Phalakarn**, and V. Suppakitpaisarn. “Parallelized Side-Channel Attack Resisted Scalar Multiplication Using q -Based Addition-Subtraction k -chains,” Proc. of the 4th In-

ternational Symposium on Computing and Networking (CANDAR 2016), pp. 140–146.

PATENTS

K. Phalakarn, T. Takisaka, T. Haas, and I. Hasuo. “System Optimal Control Device, System Optimal Control Method, and Program,” U.S. Patent 12 360 519, July 15, 2025.

OTHER
PUBLICATIONS

K. Phalakarn, K. Phalakarn, S. Prasitjutrakul, and S. Sinthupinyo. “Python 101,” Textbook for 2110101 Computer Programming course (in Thai), 2017.

PROFESSIONAL
SERVICES

- Reviewer: FMCAD 2026, CONCUR 2026, ATVA 2023
- Artifact Evaluation Committee: ATVA 2026, CAV 2026, FM 2026, VMCAI 2026
- Session Chair: ATVA 2025

SKILLS

- Programming Languages: Python, C/C++, Java; some experiences with PRISM model checker, R, VHDL, Verilog, OpenMP API for parallel programming.
- Languages: Thai (native), English (fluent), Japanese (beginner).